

AURORA: A Multi-Objective Reinforcement Learning Framework for Adaptive Cognitive State Optimization in Virtual Reality

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Abstract—Consumer-grade virtual reality (VR) systems are increasingly used for immersive gaming, training, education, and interactive applications, yet multitasking in VR can induce cybersickness, increase cognitive load, and impair working memory. Existing visual mitigation techniques, such as dynamic field-of-view (DFOV) restriction and depth-of-field (DOF) blur, mainly target cybersickness and typically rely on fixed presets or manual tuning, which may degrade task performance and user immersive experience (UIX). This paper presents AURORA, a real-time adaptive framework for consumer VR headsets that jointly optimizes cybersickness, cognitive load, and working memory during immersive interaction. AURORA uses a multitask deep learning model to estimate user cognitive states from streaming eye- and head-tracking signals and a multi-objective reinforcement learning controller to dynamically select visual mitigation techniques and adjust their intensities. A human-in-the-loop personalization module further adapts the policy using in-session comfort ratings and post-session feedback. We evaluate AURORA on a consumer-grade HTC Vive Pro Eye headset with 47 participants in a VR maze multitasking task. Results show that AURORA reduces cybersickness, balances cognitive load, and improves working-memory performance compared with static techniques, while preserving immersion and task performance. Approximately 89% of participants rated AURORA as more effective, less demanding, and more immersive. These results demonstrate the potential of adaptive, user-aware optimization for improving comfort and usability in consumer VR systems. The implementation is available at: <https://github.com/dependable-cps/AURORA>.

Index Terms—Reinforcement learning, virtual reality, cybersickness, cognitive load, working memory, personalization.

I. INTRODUCTION

The increasing adoption of consumer-grade virtual reality (VR) headsets, including Meta Quest, HTC Vive, Apple Vision Pro, and Pico, has enabled immersive applications in healthcare, education, military training, entertainment, and interactive consumer services [1]–[3]. Many of these applications require users to perform multitasking activities such as navigating complex environments, following auditory instructions, tracking objects, and making time-sensitive decisions [2], [4]–[6]. Such tasks place substantial demands on perceptual and cognitive resources, often leading to cybersickness, elevated cognitive load, and degraded working memory [7], [8]. Among these factors, cybersickness has received the most attention

because of its immediate symptoms and its role as a major barrier to broader consumer VR adoption [3], [8]–[11]. To mitigate cybersickness, prior studies have explored visual techniques such as dynamic field-of-view (DFOV) restriction [12], [13], dynamic Gaussian blur (DGB) [14], depth-of-field (DOF) blur [15], [16], and rest frames (RFs), including virtual noses and peripheral teleportation [17], [18]. Although effective in reducing discomfort, these techniques modify the user's visual perception and may negatively affect other cognitive states [3], [8], [19]. For example, a restricted field of view can force additional head movements, increasing cognitive load and impairing working memory [19], [20], while blur-based methods can reduce visual interpretability and increase cognitive effort [21]. Moreover, their effectiveness depends not only on the selected technique but also on its intensity and triggering strategy. Static or manually tuned settings, such as fixed DFOV aperture size or blur intensity, may reduce cybersickness but degrade task performance, spatial awareness, and user immersive experience (UIX). Existing approaches largely optimize cybersickness in isolation and do not jointly account for cognitive load and working memory, nor do they continuously adapt visual technique selection and intensity based on real-time user states [9], [10], [18], [22]–[24].

A plethora of recent works have also used machine learning (ML) and deep learning (DL) models to predict VR user states from sensor data, including eye tracking, head tracking, and physiological signals [4], [25]–[34]. However, most models are trained separately for individual states, limiting their ability to capture shared dependencies among cybersickness, cognitive load, and working memory. In addition, these works are primarily predictive and do not use the estimated states to drive adaptive mitigation. More recent systems, such as those in [9], [24], predict cybersickness severity and map explanations to predefined visual techniques, but they remain single-objective, rely on fixed-intensity presets, and do not optimize multiple cognitive states simultaneously. These limitations motivate a real-time framework that jointly predicts user states and adaptively controls visual mitigation strategies to improve comfort, task performance, and UIX in consumer VR.

To address this gap, we propose *AURORA*, a multi-objective reinforcement learning (RL) framework for adaptive visual optimization in consumer VR. AURORA integrates a multitask learning (MTL)-enabled DL model that predicts user cognitive states from streaming eye- and head-tracking data, and an RL agent that uses these predictions to dynamically select visual techniques from a library including DFOV, DGB, DOF, and RF and adaptively adjust their intensity, enabling context-aware policies that optimize overall UIX. In addition, AURORA incorporates a human-in-the-loop (HIL) fine-tuning process that

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uses in-session comfort ratings and post-session feedback to personalize the learned policy to individual tolerance profiles. In summary, the main contribution of this paper are as follows.

- We develop an MTL-enabled Transformer model that jointly predicts cybersickness severity, physical cognitive load, mental cognitive load, and working-memory levels from streaming VR eye- and head-tracking data. The model is trained and validated using the open-source *VRWalking* dataset [4] and achieves prediction accuracies of 93%, 88%, 94%, and 90%, respectively.
- We formulate adaptive VR visual mitigation as a multi-objective RL problem and train a Proximal Policy Optimization (PPO) agent [35] in domain-randomized VR maze simulations [36]. The agent learns to select visual techniques and tune their intensities (e.g., aperture size, blur intensity, focal distance) based on predicted user states and motion context.
- We design an HIL personalization method that constructs a user-specific preference vector from comfort ratings and post-session feedback, then uses it to refine reward weights and technique-specific intensity bounds during fine-tuning.

We validate AURORA with 47 participants in a VR maze multitasking study on a consumer-grade HTC Vive Pro Eye headset. Results show that AURORA reduces cybersickness, balances cognitive load, and improves working-memory performance compared with static and rule-based methods, while preserving immersion and task performance. Approximately 89% of participants reported superior comfort and usability. To the best of our knowledge, AURORA is the first framework to formulate VR user cognitive-state optimization as a multi-objective RL problem that jointly adapts visual technique selection, intensity control, and HIL personalization. To support reproducibility, we release AURORA as an open-source framework that includes the VR simulation, RL training pipeline, DL-based prediction model, and HIL personalization module: <https://github.com/dependable-cps/AURORA>

II. BACKGROUND

In this section, we review the main research directions related to AURORA, including user cognitive state prediction, adaptive visual techniques, reinforcement learning (RL) in VR, and the relationships between user states and visual adaptation.

a) User Cognitive-State Prediction: User cognitive states, such as cybersickness, cognitive load, and working memory, characterize how users perceive, process, and respond to VR environments during immersion [4], [30]. These states jointly influence comfort, task performance, and user immersive experience (UIX) [2], [7], [8]. Cybersickness commonly appears as dizziness, eye strain, and nausea caused by visual-vestibular mismatch [11]; cognitive load reflects the demand placed on limited cognitive resources [7]; and working memory enables temporary storage and manipulation of task-relevant information [8]. Existing measurement approaches include subjective scales, such as the Simulator Sickness Questionnaire (SSQ) [37], Fast Motion Sickness (FMS) scale [28], NASA Task Load Index (NASA-TLX) [4], and Digit Span Task (DST) [8], as well as objective signals from eye tracking,

TABLE I
COMPARISON OF PRIOR WORK WITH OUR PROPOSED AURORA FRAMEWORK ACROSS USER COGNITIVE STATE (UCS) PREDICTION, ADAPTATION, AND JOINT OPTIMIZATION OF UCS (JUCS).

Literature Review	UCS Prediction			UCS Adaptation			JUCS
	CS	CL	WM	CS	CL	WM	
Kundu et al. [25]	✓	×	×	×	×	×	×
Jeong et al. [38]	✓	×	×	×	×	×	×
Qi et al. [27]	✓	×	×	×	×	×	×
Zhu et al. [33]	✓	×	×	×	×	×	×
Wei et al. [29]	×	✓	×	×	×	×	×
Gao et al. [31]	×	✓	×	×	×	×	×
Pavel et al. [26]	×	✓	×	×	×	×	×
Setu et al. [32]	×	✓	✓	×	×	×	×
Kundu et al. [30]	✓	✓	✓	×	×	×	×
Fernandes et al. [12]	×	×	×	✓	×	×	×
Hussain et al. [15]	×	×	×	✓	×	×	×
Kelly et al. [13]	×	×	×	✓	×	×	×
Nie et al. [18]	×	×	×	✓	×	×	×
Islam et al. [10]	✓	×	×	✓	×	×	×
Hadadi et al. [34]	✓	×	×	✓	×	×	×
Chiossi et al. [22]	×	×	✓	×	×	✓	×
Matam et al. [23]	×	✓	×	×	✓	×	×
Kundu et al. [9], [24]	✓	×	×	✓	×	×	×
Our proposed	✓	✓	✓	✓	✓	✓	✓

head tracking, heart rate (HR), and galvanic skin response (GSR) [4], [28]. More recently, ML/DL methods have been used to automatically infer these states from multimodal VR sensor data [4], [25]–[32], [38]. For example, Gao et al. [31] used LightGBM to predict cognitive load from eye-tracking data, while Qi et al. [27] used an LSTM model to predict cybersickness severity. Other studies predicted multiple cognitive states using LSTM, GRU, or Transformer models [30], [32], but typically trained separate models for each state and did not use the predictions for adaptive optimization. In contrast, AURORA employs a unified MTL-enabled DL model to jointly predict multiple cognitive states in real time and to guide UIX optimization.

b) Visual Techniques for Adaptation: Visual adaptation techniques have been widely studied to improve comfort in VR [10], [12], [13], [15], [17], [24], [34]. These techniques include dynamic field-of-view (DFOV) restriction, depth-of-field (DOF) blur, dynamic Gaussian blur (DGB), and rest frames (RFs). Early approaches, such as subtle FOV narrowing during rapid motion [12] and DOF blur for reducing visual conflict [15], generally relied on fixed intensities or manual activation, limiting their ability to respond to changing user states. Later closed-loop systems used predicted cybersickness to trigger visual interventions [10], [24]; however, they still focused primarily on cybersickness and often applied predefined, fixed-intensity techniques. Although a few works have targeted other states, such as cognitive load [22], [23], prior methods do not jointly optimize cybersickness, cognitive load, and working memory. AURORA addresses this limitation by adaptively selecting visual techniques and continuously tuning their intensities in response to real-time cognitive-state feedback.

c) Reinforcement Learning in VR: RL has recently gained attention for adaptive and personalized VR/XR systems because it can learn context-aware policies from feedback [35], [39]–[41]. For example, Lee et al. [41] used multi-agent RL to dynamically switch redirected-walking strategies, Pelosi et al. [39] applied RL to personalize rehabilitation trajectories,

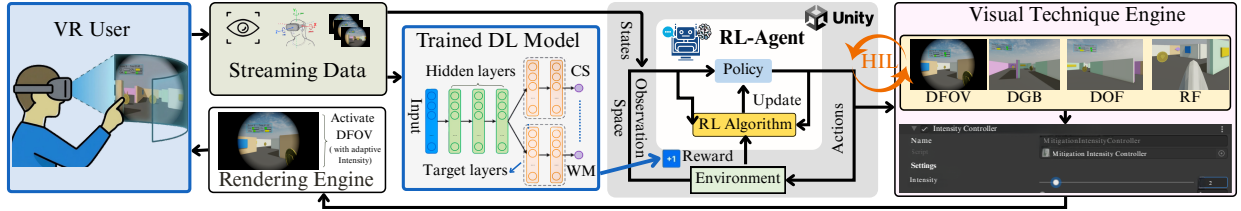


Fig. 1. Overview of the proposed AURORA framework. The MTL-enabled DL model predicts users' targeted cognitive states from streaming VR data, which shapes the RL agent's reward. The RL agent then selects the most suitable visual technique and its intensity via the adaptive visual technique engine, while a HIL fine-tuning process further personalizes the policy.

and Gehrke et al. [40] demonstrated RL-based adaptive haptic control using explicit ratings and neural feedback. These studies show the promise of RL for user-centered adaptation, but they do not address joint cognitive-state optimization. AURORA fills this gap by formulating adaptive VR visual mitigation as a multi-objective RL problem that selects techniques and tunes intensities to jointly regulate cybersickness, cognitive load, and working memory.

d) *Relationship Among User Cognitive States and Visual Techniques:* Cybersickness, cognitive load, and working memory are interdependent because they draw on shared perceptual and cognitive resources [7]. VR multitasking further strains these resources by requiring users to process continuous sensory input, maintain spatial orientation, and perform concurrent tasks [2]. Prior studies show that visually induced motion sickness can impair memory and multitasking performance [7], [8], [42], while workload can either reduce or worsen cybersickness depending on task demand and context [5], [6], [43]. Visual adaptation techniques also affect these states differently depending on their type and intensity [8], [17], [19]–[21]. For instance, DFOV and DGB may reduce cybersickness but increase cognitive effort by limiting or degrading visual information [19]–[21]; DOF can alter depth perception; and RF effectiveness may vary under high-motion conditions [17]. These findings indicate that both technique selection and intensity control should account for joint effects across multiple user states rather than optimizing cybersickness alone.

III. AURORA FRAMEWORK

This section presents the proposed AURORA framework, that jointly optimizes users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory capabilities.

A. System Overview

AURORA operates as a closed-loop system within a VR environment, as illustrated in Figure 1. The framework consists of an MTL-enabled DL model, an RL agent, a Visual Technique Engine, and a HIL personalization process. During VR immersion, the MTL-enabled DL model continuously processes streaming eye- and head-tracking data to predict user cognitive states (e.g., cybersickness severity, cognitive physical load level, cognitive mental load level, and working memory capabilities) in real time. These predictions shape a multi-objective reward signal that guides the RL agent to jointly optimize users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory.

At each decision step, the agent receives the predicted user states, along with the currently active visual technique (DFOV, DGB, DOF, and RF) and its intensity (e.g., FOV, blur, etc.), and selects the most suitable technique and intensity for real-time application. Furthermore, user feedback collected during and after each session is combined with the predicted user cognitive states to build a personalized preference vector that drives HIL-based offline policy fine-tuning between sessions.

B. MTL-Enabled DL Model for RL Reward Shaping

For an effective visual technique (e.g., DFOV, DGB, etc.) selection with accurate intensity, we require real-time predictions of the targeted user's cognitive states (e.g., cybersickness, cognitive load, and working memory), since these predictions directly shape the reward used for RL policy learning. Accurate predictions help the RL agent relate its visual-technique actions to their effects on user state, thereby supporting more stable learning and better long-term policy behavior. Therefore, we develop an MTL-enabled DL model that jointly predicts cybersickness severity, cognitive physical load, cognitive mental load, and working memory levels within a single unified architecture. Specifically, we use a SOTA Transformer-based DL model, which has been increasingly adopted in recent VR user-state prediction studies [24], [26], [30]. The Transformer model includes a positional encoding (PE) layer for input projection and temporal encoding, a Transformer encoder with 64-dimensional multi-head self-attention, residual add-and-norm layers, and a 256-dimensional feed-forward MLP layer (256 dimensions) to capture latent temporal features from VR sensor data, followed by task-specific output layers for severity prediction. Unlike single-task models, which are trained independently for each target state and produce one prediction at a time [30], [32], our proposed MTL-enabled DL model is trained jointly on all target states using a shared backbone with task-specific prediction heads. This shared design is particularly suitable for our RL reward-shaping pipeline because these user states are closely coupled during VR immersion. We train the model on the open-source VRWalking dataset [4] using task-specific categorical cross-entropy losses. Although each task has a different number of class labels, categorical cross-entropy is applied separately to each task's output, and the overall MTL objective is jointly optimized within the shared architecture. For instance, in the cybersickness prediction task, have four severity classes (e.g., none, low, medium, and high), while cognitive load and working memory tasks have three class labels (e.g., low, medium, and high). Thus, the same loss function is used across all tasks, while each task retains its own label space within the shared MTL framework. Because the RL agent requires

continuous feedback rather than only discrete class labels, we convert each softmax output into a normalized continuous severity score, as defined in Equation 1 in Section III-C, so that all outputs are mapped to $[0, 1]$ regardless of the number of classes. These normalized severity scores are then used to compute the multi-objective reward in Equation 2. For deployment, we export the trained model to ONNX and integrate it into the Unity runtime.

C. Multi-Objective RL Agent

The RL agent in AURORA is designed to jointly optimize users' cognitive states, which we formalize this process as a Markov Decision Process (MDP), represented by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

State ($s_t \in \mathcal{S}$): The state encodes multimodal features that are informative for the targeted user's cognitive states (e.g., cybersickness (CS), cognitive mental load (CML), cognitive physical load (CPL), and working memory (WM)). Specifically, it includes locomotion signals (e.g., walking speed and angular velocity), head-tracking signals (e.g., position, rotation, and rotational velocity), eye-tracking signals (e.g., pupil diameter, gaze direction, and fixation stability), visual motion cues (e.g., optic flow magnitude), and task context (e.g., turning vs. straight walking). To capture temporal dependencies, the previously applied visual technique and its intensity are also included.

Action ($a_t \in \mathcal{A}$): At each timestep t , the agent selects a composite action $a_t = (M_t, I_t)$, where $M_t \in \{\text{DFOV, DGB, DOF, RF}\}$ specifies one visual technique from our visual technique library and $I_t \in [0, 1]$ denotes the corresponding normalized intensity. Only one visual technique is applied at each timestep.

Transition dynamics (P): The transition function $P(s_{t+1} | s_t, a_t)$ describes how the user's multimodal state changes in response to different visual-technique actions, including changes in motion patterns, head and eye behavior, visual motion cues, and targeted user states.

Reward (R): The reward is derived from the predictions of the DL-based MTL model (see Section III-B) for the targeted users' cognitive states (e.g., CS, CML, CPL, and WM), unlike prior single-objective formulations that focus only on reducing cybersickness [9], [10], [15], [24]. Let $\hat{s}_{k,t}$ denote the normalized continuous prediction value for user state k at time t , computed as:

$$\hat{s}_{k,t} = \frac{1}{K_k - 1} \sum_{i=0}^{K_k-1} i \cdot p_{k,i,t} \quad (1)$$

where K_k is the number of classes for users' cognitive state k ($K_{\text{CS}} = 4$ and $K_{\text{CPL}} = K_{\text{CML}} = K_{\text{WM}} = 3$), and $p_{k,i,t}$ is the softmax probability of class i at time t from the MTL-enabled DL model in Section III-B. The factor $(K_k - 1)^{-1}$ normalizes all user-state predictions to $[0, 1]$ regardless of class count so that each state contributes on a comparable scale to the reward. The reward at each timestep is then given by:

$$r_t = \sum_{k \in \mathcal{K}} \alpha_k d_k (\hat{s}_{k,t-1} - \hat{s}_{k,t}) - \lambda_m I_t - \lambda_\Delta |I_t - I_{t-1}| - \lambda_{\text{switch}} \cdot \mathbb{I}[M_t \neq M_{t-1}] + \lambda_{\text{stab}} \cdot (1 - |I_t - I_{t-1}|) \quad (2)$$

where $\mathcal{K} = \{\text{CS, CPL, CML, WM}\}$, α_k is the weight assigned to the user's cognitive state k , and d_k is a direction term that is $+1$ for states where a lower value is preferable (CS, CPL, and CML) and -1 for working memory, where a higher value is preferable. The penalty terms regulate excessive visual-technique intensity (λ_m), abrupt intensity changes (λ_Δ), and frequent technique switching (λ_{switch}), while the stability bonus (λ_{stab}) encourages smooth transitions. This formulation enables the agent to discover trade-offs across targeted user states. This formulation enables the agent to discover trade-offs between competing user cognitive states. For example, visual techniques (e.g., DFOV, DGB, etc.) designed to reduce cybersickness may also increase cognitive load and degrade working memory [7], [8], [19]. The agent therefore learns to select intensity levels that balance these user cognitive states rather than optimizing any single state in isolation.

Discount factor (γ): The parameter $\gamma \in [0, 1]$ controls how the RL agent balances immediate and long-term effects of its actions. A moderate value encourages the agent to consider the cumulative impact of visual technique adaptation on user states over time. We train the RL agent using Proximal Policy Optimization (PPO) [35]. We select PPO because it can effectively handle our mixed action space, which involves both discrete visual-technique selection and continuous intensity control, while maintaining stable policy updates through its clipped surrogate objective, which is critical in our setting where reward signals from the prediction model carry inherent noise. Training is performed in domain-randomized VR maze environments with variations in layout, motion dynamics, visual flow, and walking speed to improve policy robustness. PPO updates are carried out over mini-batches and multiple epochs, guided by the multi-objective reward parameters (α_k , λ_m , λ_Δ , λ_{switch} , λ_{stab}), which are tuned through preliminary experiments, as discussed in section IV.

D. Adaptive Visual Technique Engine

During runtime, the RL policy observes the current cognitive state and outputs an action $a_t = (M_t, I_t)$, where M_t denotes the selected visual technique and I_t denotes its intensity. This action is passed to the adaptive visual technique engine, which includes four SOTA visual techniques: DFOV [12], DGB [14], DOF [15], and RF (virtual nose [17]). As discussed in Section II, these techniques affect user cognitive states differently across motion contexts and individual users, so no single fixed strategy performs well in all VR scenarios. The RL agent, therefore selects the most suitable visual technique and its intensity based on the current context and predicted user cognitive states. Each decision is then evaluated through the multi-objective reward in Equation 2, enabling the policy to jointly optimize users' cognitive states by reducing cybersickness while balancing cognitive load and improving working memory during VR immersion. We implement DFOV using Unity's Tunneling Vignette Controller, where the aperture fraction narrows the visible field from 120° to 60° as intensity increases. DGB is implemented using Unified Universal Blur within Unity's Universal Render Pipeline (URP), where the intensity parameter controls blur strength ($\sigma = 0$ to 15). DOF is implemented through URP's post-processing depth buffer

TABLE II
PPO TRAINING AND HIL FINE-TUNING SETTINGS FOR AURORA.

HP	Value	HP	Value
Learning rate (η)	1.0×10^{-4}	α_{CS}	0.50
Clipping ratio (ϵ)	0.2	α_{CPL}	0.15
Entropy coeff. (β)	1.0×10^{-2}	α_{CML}	0.15
Discount factor (γ)	0.995	α_{WM}	0.10
Mini-batch size	512	λ_m	0.05
Buffer size	10240	λ_Δ	0.10
Num. epochs per update (K)	5	λ_{switch}	0.08
Full training steps	5.0×10^5	λ_{stab}	0.05
HIL fine-tuning steps	5.0×10^4	Hidden units	256

with Bokeh mode, allowing focal distance and blur strength to be adjusted continuously. For RF, following the virtual nose approach [17], we attach a nose model to the user's view and allow the RL agent to adaptively control its width based on the current user cognitive state. Although we focus on these four techniques, the engine can be extended to include other adjustable visual techniques, such as color manipulation [3], peripheral teleportation [18] and snap turning [13], as long as their effects can be modeled within the same RL framework.

E. HIL-Based Personalization of Visual Techniques

The baseline AURORA policy is designed for a general population, but tolerance to visual techniques and preferences for their intensity can vary substantially across users [3]. For example, some users may tolerate stronger blur but find FOV restriction disruptive, whereas others may prefer minimal visual alteration even if it provides less relief. To account for this variability, we incorporate a HIL personalization process that fine-tunes the policy using user-specific feedback collected during and after immersion. During VR sessions, participants periodically report their current user cognitive state through verbal feedback, while the MTL-enabled DL model concurrently predicts the remaining targeted user cognitive states in real time (Section III-B). Together, these reported and predicted states form a per-technique response profile that captures whether a given visual technique, at its applied intensity, improves the user's overall condition or introduces undesirable side effects. After each session, participants also complete a structured preference questionnaire that captures technique-specific comfort, perceived visual disruption, preferred intensity level, and overall technique ranking. We combine these feedbacks to construct a user-specific preference vector $\mathbf{z}$, which encodes personalized adjustments to the reward weights α_k in Equation 2, technique-specific intensity bounds, and relative preferences over visual techniques. We then use $\mathbf{z}$ to refine the reward parameters and impose technique-specific constraints during offline policy fine-tuning with PPO between sessions, after which the personalized policy is deployed in the next evaluation session. For example, if a user reports that high-intensity DGB is uncomfortable while DFOV remains acceptable, the fine-tuned policy reduces the allowable DGB intensity range and shifts preference toward DFOV, while continuing to optimize across all targeted user states.

IV. DATASETS & EXPERIMENTAL SETUP

This section details the dataset used for training and evaluation, Experimental Setup, and system performance analysis.

TABLE III
MTL-ENABLED TRANSFORMER MODEL PERFORMANCE FOR DIFFERENT USER COGNITIVE STATES (UCS) ACROSS SEVERITY LEVELS (NONE (N), LOW (L), MEDIUM (M), AND HIGH (H)), WHERE CPL, CML, AND WM HAVE THREE LEVELS (L, M, H) AND A DENOTES ACCURACY.

UCS	A(%)	Precision (%)				Recall (%)				F1-score (%)			
		N	L	M	H	N	L	M	H	N	L	M	H
CS	93	98	90	85	96	96	88	82	95	97	89	83	95
CPL	88	—	90	82	92	—	87	79	90	—	88	80	91
CML	94	—	95	89	96	—	93	87	94	—	94	88	95
WM	90	—	92	84	93	—	89	82	91	—	90	83	92

Dataset: We use the open-source VRWalking dataset [4], which includes eye-tracking, head-tracking, and physiological data from 39 participants (23 male, 16 female; mean age 25.67, SD = 7.22) navigating a virtual maze by real walking for 15 minutes. The dataset provides eye-tracking features (e.g., gaze direction, pupil position, etc), head-tracking features (e.g., position, rotation, etc.), and physiological signals (e.g., HR, GSR) with ground-truth labels for six multitasking metrics: cybersickness, cognitive mental load, cognitive physical load, working memory, attention success rate, and attention reaction time. In this work, we specifically focus on cybersickness, cognitive mental load, cognitive physical load, and working memory to train the DL-based MTL prediction model and shape the RL reward. Following prior studies [9], [25], cybersickness is categorized into four severity levels (none, low, medium, high), while the remaining user cognitive states retain the original three levels (low, medium, and high) as provided in the VRWalking dataset [4], [32]. These discretized severity labels serve as ground truth for training the MTL-enabled DL model (Section III-B). It is worth mentioning that we rely solely on integrated sensor data (e.g., eye and head tracking) rather than external physiological signals (e.g., HR, GSR), as these require tethering to the user's body and are prone to noise [24], [28].

Data Preprocessing: Before training AURORA and fine-tuning AURORA-HIL, we preprocess the collected multimodal data to ensure consistency across sensor modalities. We first remove outliers beyond three standard deviations from the continuous multimodal features using z-score analysis [44]. We then denoise the time series using exponential smoothing [45], $\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t$, where $\hat{y}_{t+1}$ is the forecasted value at time $t + 1$, $\hat{y}_t$ is the previous smoothed estimate, and α is the smoothing factor between 0 and 1. We set $\alpha = 0.8$ based on preliminary experiments to balance noise reduction and signal fidelity. Finally, we normalize each continuous feature as $X_{\text{normalized}} = (X - \mu)/\sigma$, where $X_{\text{normalized}}$ represents the normalized sample, and μ and σ denote the mean and standard deviation of the corresponding feature.

Experimental Setup: To validate the proposed AURORA framework, we use a VR maze simulation consistent with the environment in the VRWalking dataset [4], ensuring compatibility between the training data distribution and the deployment setting. The system is developed in Unity and rendered using the URP. AURORA is deployed on a consumer-grade head-mounted display (HMD) HTC Vive Pro Eye with integrated eye tracking and head tracking, a resolution of 2880×1600 pixels per eye, and a refresh rate of 90 Hz. The complete system, including RL training and model inference, runs on a workstation equipped with an Intel Core i9-10900X

CPU, 128 GB of RAM, and an NVIDIA RTX 3090 GPU.

a) Virtual Environment and Apparatus: To ensure compatibility between training and deployment, we replicate a VR maze environment consistent with the VRWalking dataset [4], on which both the MTL-enabled Transformer model and the RL agent are trained. The maze is deployed within a $5\text{ m} \times 5\text{ m}$ physical walking area, with each platform measuring approximately $0.5\text{ m} \times 0.5\text{ m}$. Maze layouts are generated procedurally using a depth-first search-based algorithm, producing connected structures with varied paths across trials. The environment includes colored walls, a cold sunset sky, and no ceiling, as shown in Figure 4a. The HTC Vive Pro Eye streams real-time eye- and head-tracking data to the MTL-enabled Transformer model and the RL policy during execution.

b) RL Training Setup and Hyperparameter Selection: The RL agent is implemented using the Unity ML-Agents Toolkit [46] and trained with PPO [35]. The MTL-enabled Transformer model is implemented in PyTorch 2.3. To improve generalization across users and environments, we train the RL agent in domain-randomized maze environments [36] with variations in layout complexity, corridor width, turn frequency, and visual appearance. This exposes the agent to diverse motion and contextual conditions during training. At each decision step, the agent receives a sliding window of recent sensor observations and outputs both a visual technique and its corresponding intensity. The PPO hyperparameters are summarized in Table II. We initialize them based on the original PPO formulation [35] and prior large-scale studies on on-policy RL training [47], and then refine them through grid search over preliminary training runs [48]. During this RL agent training process, we jointly tune the reward weights α_k in Equation 2 and the penalty coefficients ($\lambda_m, \lambda_\Delta, \lambda_{\text{switch}}, \lambda_{\text{stab}}$). The final reward configuration assigns the largest weight to cybersickness ($\alpha_{\text{CS}} = 0.50$), followed by cognitive physical load and cognitive mental load ($\alpha_{\text{CPL}} = \alpha_{\text{CML}} = 0.15$), and working memory ($\alpha_{\text{WM}} = 0.10$). This weighting reflects the greater optimization pressure required to reduce cybersickness while balancing cognitive load and improving working memory. Training is stopped when the cumulative reward on a held-out set of unseen maze configurations no longer improves.

System Benchmarks: For real-time deployment on a consumer-grade VR system, both the DL-based prediction model and the RL policy must execute within the VR frame budget. At 90 Hz, each frame allows approximately 11.1 ms for all processing, including scene rendering, input handling, and AURORA's inference pipeline. In our implementation, the MTL-enabled DL model runs via Unity Sentis on the GPU in 1.8 ms, the RL policy executes on the CPU Burst backend in 0.3 ms, and the visual technique engine render pass completes in 0.9 ms, yielding a total pipeline overhead of approximately 3.0 ms. This is less than 27% of the per-frame budget, remaining within acceptable limits for SOTA real-time VR settings [24].

V. TECHNICAL EVALUATION RESULTS

This section presents the technical evaluation of the AURORA framework, including the MTL-enabled DL model for

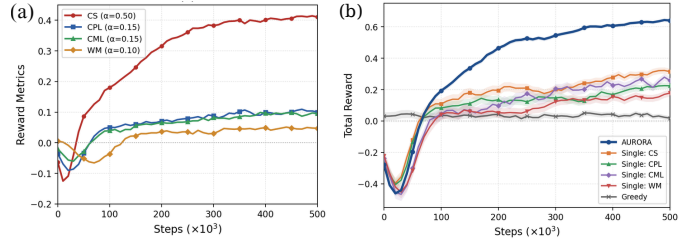


Fig. 2. Reward evolution and policy comparison during RL training. (a) Per-objective reward contributions of AURORA for cybersickness (CS) reduction, cognitive physical load (CPL) balancing, cognitive mental load (CML) balancing, and working-memory (WM) improvement. (b) Mean total multi-objective reward of AURORA compared with four single-objective RL agents (CS, CPL, CML, WM) and a greedy policy baseline.

RL reward shaping, the reward metric evolution during RL training, and the learned action distribution analysis.

DL Models for RL Reward Shaping: Before integrating the DL-based prediction models into RL training, we evaluate their predictive performance for the targeted user cognitive states. For AURORA, we use the proposed MTL-enabled DL model, which jointly predicts cybersickness, cognitive physical load, cognitive mental load, and working memory using a shared backbone with task-specific output heads. Table III summarizes the accuracy, precision, recall, and F1-score across all four targeted user cognitive states. Overall, the MTL-enabled model achieves consistently strong performance, with classification accuracies of 93%, 88%, 94%, and 90% for cybersickness, cognitive physical load, cognitive mental load, and working memory, respectively. Furthermore, the model achieves an inference time of approximately 1.8 ms per frame, which is within the real-time budget for 90 Hz rendering [24]. To further evaluate AURORA against single-objective RL agents, we also train four single-task DL models following prior VR user-state prediction studies [28], [49]–[51]. Each model is trained to predict only one cognitive state and is paired with a corresponding single-objective RL agent. The single-task DL models achieve accuracies of 95%, 89%, 93%, and 90% for cybersickness, cognitive physical load, cognitive mental load, and working memory, respectively. This setup enables a direct comparison between the proposed joint multi-objective reward shaping and single-state reward shaping under the same environment and action space.

Reward Metric Evolution: Figure 2 shows the reward evolution of AURORA and its comparison with alternative policy designs. As shown in Figure 2(a), the reward components initially remain negative while the PPO agent explores different visual techniques and intensity levels. At the beginning of training, the cybersickness term drops the most because stronger mitigation actions may reduce cybersickness but can temporarily increase cognitive load or reduce working memory. As training progresses, the cybersickness reward rises sharply, while the cognitive physical load, cognitive mental load, and working memory rewards improve more gradually, indicating that the agent learns to balance cybersickness reduction with cognitive load and working-memory preservation. In the final phase, all reward terms stabilize, suggesting convergence to a stable multi-objective policy. Figure 2(b) further compares AURORA with four single-objective RL

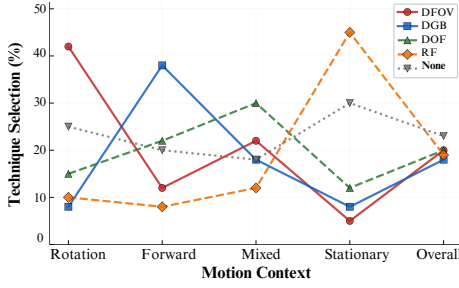


Fig. 3. Distribution of RL-selected visual techniques across different VR motion contexts.

agents trained separately for cybersickness, cognitive physical load, cognitive mental load, and working memory, as well as a greedy policy. All policies are evaluated using the full multi-objective reward in Eq. (2). The single-objective agents achieve lower total reward because optimizing one cognitive state alone can negatively affect the remaining states, whereas the greedy policy selects the action with the highest immediate reward but cannot account for the long-term effects of visual adaptation. Overall, AURORA achieves the highest total reward, demonstrating the benefit of jointly optimizing all cognitive-state objectives through PPO-based policy learning.

Action distribution analysis: We analyze the RL agent’s action selections over held-out evaluation episodes using the trained policy described in Section IV. At each decision step, we record the selected visual technique, group actions by motion context (e.g., rotation, forward, mixed, and stationary), and compute the percentage of selections for each technique within each context, as shown in Figure 3. The resulting distribution reveals clear context-dependent behavior. During rotational motion, DFOV is selected most often (42%), consistent with its role in suppressing peripheral optical flow induced by head turns. During forward walking, DGB dominates (38%), reflecting its effectiveness in reducing translational flow during locomotion. In mixed movement contexts, DOF is preferred (30%), suggesting that depth-selective defocusing is better suited to combined rotational and translational motion. During stationary periods, RF peaks at 45%, providing a low-cost visual anchor when cybersickness risk is relatively low. Notably, the None action accounts for 20–30% of selections across contexts, indicating that the agent has learned to withhold visual adaptation when predicted user states remain within acceptable ranges.

VI. USER STUDY DESIGN AND EVALUATION

This section presents the user study design and evaluation results to show the effectiveness of our proposed framework.

A. User Study Design on AURORA

1) *Hypotheses:* This study aims to evaluate the effectiveness of AURORA, which uses multi-objective RL to jointly optimize users’ cognitive states by dynamically selecting the most suitable visual technique and adaptively adjusting its intensity to reduce cybersickness, balance cognitive load, and improve working memory in VR. Based on this objective, we formulate the following hypotheses to compare AURORA against the conventional method (where the visual techniques are applied using static, fixed, or predefined rules):

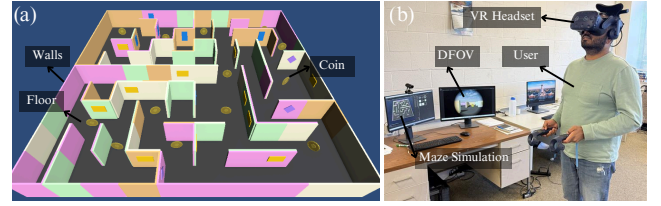


Fig. 4. (a) Top-down view of the procedurally generated VR maze environment. (b) Participant immersed in the proposed AURORA framework.

- H1:** AURORA with adaptive visual techniques will reduce cybersickness, balance cognitive load, and improve working memory compared to conventional methods.
- H2:** Participants will experience a better UX and a higher sense of presence in the virtual environment when using AURORA compared to the conventional methods.
- H3:** HIL-personalized AURORA will further improve outcomes compared to non-personalized AURORA.

2) *Participants, Study Task, and Procedure:* Our user study involved 47 volunteer participants (26 male, 21 female) aged from 18 to 39 (mean age 27.2, SD 3.26). The participants were Asian, American, Black, and Hispanic. Participants reported no vestibular dysfunction and were not taking anti-motion sickness medication. Thirty-two of the 47 participants reported prior VR experience, whereas the rest stated they had never used VR equipment. Our user study was approved by the Institutional Review Board (IRB) at the authors’ university. All participants completed an IRB-approved consent form and a demographic questionnaire (e.g., age, gender, ethnicity, prior VR experience, baseline SSQ, etc.). Participants were briefed on the study procedures and informed of their right to withdraw at any time if they felt too uncomfortable. They then entered the VR study using the HTC Vive Pro Eye headset and controllers and completed a tutorial that outlined the study tasks. Further details are provided below.

Warm-up Phase: Before the experiment, participants were briefed on the study procedures and completed a virtual tutorial introducing the tasks: *coin collection* for cognitive load measurement and *digit span recall* for working memory assessment. They first engaged in a practice session to become familiar with real walking locomotion in the maze and the study tasks. The tutorial also ensured that participants could clearly hear instructions through the HMD’s integrated headphones. During the tutorial, participants practiced the *coin-collection task*, where coins were placed at random locations throughout the maze corridors and collected by walking close to them. They also practiced the *digit span recall task*. To assess working memory, we used a VR-based version of the DST [52]. Participants received a randomly generated five-digit sequence every 30 seconds, were given 20 seconds to memorize it, and 10 seconds to recall the digits in the same order during the VR simulation. For example, if given the sequence {6 – 2 – 1 – 9 – 5}, participants were expected to recall it in the same order. Working memory performance was computed as the percentage of correctly recalled digits, following [4]. Participants were instructed to verbally report their discomfort on an FMS scale from 0 to 10 (0 = no sickness, 10 = severe sickness) whenever a pre-recorded voice prompt was played every 30 seconds during the VR simulation. We

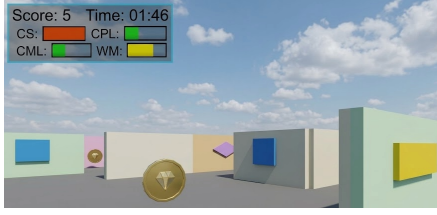


Fig. 5. User's view in the VR maze simulation without any visual technique.

used this scale to measure real-time cybersickness and to remain consistent with prior DL-based cybersickness detection studies [28]. Furthermore, we asked a single question adapted from the NASA-TLX once per minute: *How physically and mentally demanding is the task on a scale of 1 (not demanding) to 10 (very demanding)? Please provide two ratings, one for each.* to capture the participants' real-time perception of cognitive physical load and cognitive mental load while navigating the VR environment. We combined both into a single query, as repeatedly querying all user state in detail would itself increase cybersickness, elevate cognitive load, and degrade working memory. Note that these measurement scales are widely adopted in SOTA VR user cognitive state assessment studies [4], [9], [10], [24], [32].

Experiment Phase: After completing the tutorial, the VR maze simulation began and lasted 10 minutes unless participants chose to stop earlier because of discomfort. It is important to note that eye and head positions were carefully calibrated before each simulation. Upon satisfactory calibration, participants were immersed in the VR maze environment. After completing each condition, participants completed the Igroup Presence Questionnaire (IPQ) [53] and were then given a break. They were allowed to rest for as long as needed until they felt ready to continue without discomfort. The experiment concluded once participants had completed all study conditions, including the baseline condition in which no visual technique was applied. To evaluate the effectiveness of the proposed AURORA framework and compare it with conventional methods, the experiment was conducted under four conditions: Baseline, Static visual techniques, AURORA, AURORA-HIL. In the **Static visual techniques** condition, fixed-intensity visual techniques were applied, where participants experienced DFOV, DGB, DOF, and RF one by one in randomized order within the same condition, following prior studies [9], [24]. The fixed settings were selected based on prior work: a DFOV aperture of 90° [12], DGB with $\sigma = 8$ [14], DOF with a moderate gaze-contingent focal distance and blur strength [15], and RF with a moderate width and full visibility [17]. In the **AURORA** condition, the trained multi-objective RL policy automatically selected the suitable visual technique from the technique engine and adjusted its intensity in real time, as described in Section III-D. In the **AURORA-HIL** condition, the personalized RL policy was evaluated with the same participants on a separate day. This policy was fine-tuned offline using participant-specific data collected during the AURORA condition and then used to select the visual technique and intensity according to each participant's preference profile, as described in Section III-E. Participants were free to stop the experiment at any time if

they felt significantly uncomfortable. It is important to note that, to reduce order effects, the first three conditions were counterbalanced across participants. Only the AURORA-HIL condition was evaluated last on a separate day because the personalized policy requires prior participant-specific interaction data and offline fine-tuning before deployment. It is worth mentioning that prior work has shown that cybersickness symptoms can persist from hours up to several days after VR exposure [54], [55]. Therefore, the separate-day design does not necessarily eliminate residual cybersickness effects. Figure 4(a) shows the top-down view of the procedurally generated maze, Figure 4(b) shows a participant immersed in the proposed AURORA framework, and Figure 5(a) and Figure 6(a)–(d) illustrate example views of the maze under the baseline and different static visual techniques.

Post-task Interview Phase: In the post-task interview phase, participants first completed the post-SSQ and NASA-TLX. We then conducted semi-structured interviews lasting 1–2 minutes to gather participants' verbal feedback on their experiences, focusing on cybersickness, cognitive load, and working memory across conditions, using a 5-point Likert scale. Afterward, they responded to a set of follow-up statements, such as: “AURORA better balanced my discomfort, cognitive demand, and working memory than the conventional methods”, etc. We also gathered qualitative feedback through open-ended questions on participants' experiences with AURORA compared to conventional methods, beginning with open-ended questions to avoid priming. Before leaving, participants were reminded they could contact the research team if they experienced any discomfort or adverse effects. So far, no concerns have been reported from any participants.

B. User Study Results

This section compares the proposed AURORA framework against conventional methods and the baseline (without visual techniques). We first analyze the experimental results across cybersickness, cognitive load, and working memory during immersion, and then present the post-task evaluation results.

Experiment Phase: The descriptive statistics (mean M and SD) of cybersickness (CS) measured via FMS, cognitive physical load (CPL) and cognitive mental load (CML) measured via the NASA-TLX, and working memory (WM) measured via the DST across conditions are shown in Table IV. The results show that AURORA-HIL achieves the lowest CS score ($M = 1.12$, $SD = 1.02$), followed by AURORA ($M = 1.52$, $SD = 1.32$), whereas the baseline produces the highest CS level ($M = 6.30$, $SD = 2.10$). In addition, AURORA and AURORA-HIL also yield the lowest CPL and CML values and the highest WM values across all conditions, indicating that the proposed framework effectively reduces CS while balancing cognitive load and improving WM. To assess the statistical significance of these differences, we first conduct a Shapiro–Wilk test, which indicates that the collected data are not normally distributed. We therefore conduct a nonparametric Friedman test at $\alpha = 0.05$, which indicates significant differences across conditions for all four user states: CS ($\chi^2(6) = 180.53$, $p < .001$, $W = 0.640$), CPL ($\chi^2(6) = 112.78$, $p < .001$, $W = 0.400$), CML ($\chi^2(6) =$

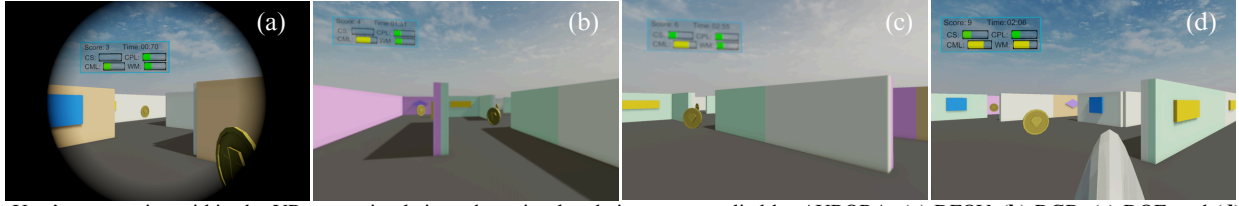


Fig. 6. User's perspective within the VR maze simulation when visual techniques are applied by AURORA: (a) DFOV, (b) DGB, (c) DOF, and (d) RF.

93.54, $p < .001$, $W = 0.332$), and WM ($\chi^2(6) = 58.88$, $p < .001$, $W = 0.209$), with Kendall's W indicating a large effect for CS, moderate effects for CPL and CML, and a small but meaningful effect for WM. Following these significant results, we conduct post-hoc pairwise Wilcoxon signed-rank tests with the Holm–Bonferroni correction to control the family-wise error rate across all pairwise comparisons within each outcome, as reported in Table VI. Note that all (p)-values reported for the post-hoc pairwise comparisons refer to Holm–Bonferroni adjusted (p)-values. The pairwise results show that all four static visual techniques significantly reduce CS relative to the baseline (all $p < .001$, $r = 0.61$ – 0.84), confirming measurable relief of cybersickness. However, their effects on CPL, CML, and WM vary and in several cases worsen relative to baseline, suggesting that static techniques reduce cybersickness without reliably balancing cognitive load or preserving working memory. In contrast, our proposed AURORA significantly reduces CS compared with all four static visual techniques (all $p < .001$, $r = 0.80$ – 0.85) and the baseline ($p < .001$, $r = 0.851$), achieving a 75.8% reduction in CS ($M = 1.52$, $SD = 1.32$). At the same time, AURORA improves all other targeted cognitive states relative to the baseline, with lower CPL ($p = .050$, $r = 0.38$), lower CML ($p < .001$, $r = 0.64$), and higher WM ($p = .020$, $r = 0.47$), as detailed in Table VI. Compared with AURORA, the static techniques impose significantly higher CML for DGB, DOF, and RF (all $p < .001$, $r = 0.56$ – 0.71), higher CPL for DFOV and RF ($p \leq .011$, $r = 0.47$ – 0.54), and lower WM for RF and DGB ($p \leq .037$, $r = 0.43$ – 0.59). These findings strongly support **H1**, confirming that AURORA, through adaptive visual-technique selection and continuous intensity control, simultaneously reduces cybersickness, balances cognitive load, and improves working memory compared with conventional methods.

Beyond AURORA's advantage over conventional methods, we further evaluate whether HIL personalization provides additional benefit. As shown in Table V, AURORA-HIL achieves an 82.2% reduction in CS from baseline ($M = 1.12$, $SD = 1.02$) and further reduces CPL by 27.4% and CML by 35.8% while improving WM by 30.0%. Compared with non-personalized AURORA, AURORA-HIL significantly improves CS ($p = .045$), CPL ($p = .009$), and WM ($p < .001$, $r = 0.80$), while CML does not differ significantly ($p = .968$), consistent with both achieving strong CML reduction relative to baseline. These findings strongly support **H3**, confirming that HIL personalization further enhances the effectiveness of AURORA across targeted cognitive states. To evaluate the proposed framework in terms of presence, we compare the IPQ presence scores of AURORA, AURORA-HIL, and the conventional methods, collected after each condition on a 7-

TABLE IV
DESCRIPTIVE STATISTICS (MEAN SD) ACROSS ALL CONDITIONS.

Condition	CS	CPL	CML	WM
Baseline	6.30 (2.10)	5.54 (1.52)	6.15 (1.77)	4.63 (1.90)
DFOV	4.35 (1.78)	5.95 (1.55)	5.09 (1.58)	5.13 (1.78)
DGB	4.80 (1.85)	4.15 (1.28)	6.49 (1.65)	4.75 (1.62)
DOF	3.90 (1.72)	5.05 (1.42)	6.50 (1.55)	4.90 (1.68)
RF	5.50 (2.05)	5.71 (1.48)	6.02 (1.68)	4.33 (1.72)
CR	1.52 (1.32)	4.77 (1.25)	4.33 (1.38)	5.79 (1.45)
CR-HIL	1.12 (1.02)	4.02 (1.18)	3.95 (1.28)	6.02 (1.22)

TABLE V
PERCENTAGE CHANGE (Δ) RELATIVE TO BASELINE. **RED VALUES** INDICATE WORSENING COMPARED TO BASELINE.

Technique	CS (%)	CPL (%)	CML (%)	WM (%)
DFOV	30.9	7.4	17.2	10.9
DGB	23.8	25.1	5.6	2.7
DOF	38.1	8.8	5.7	5.8
RF	12.7	3.0	2.1	6.5
CR	75.8	13.9	29.6	25.1
CR-HIL	82.2	27.4	35.8	30.0

point Likert scale. Because all conditions were experienced by the same participants, we apply the same repeated-measures pipeline used for the other outcomes rather than independent-samples tests. AURORA-HIL achieves the highest presence score ($M = 5.94$, $SD = 0.74$), followed by AURORA ($M = 5.61$, $SD = 0.87$), while the static techniques yield lower presence (ranging from $M = 3.57$ to $M = 4.13$) and the baseline the lowest ($M = 3.12$, $SD = 1.16$). A Shapiro–Wilk test indicates that the presence scores are not normally distributed, so we conduct a Friedman test followed by Wilcoxon signed-rank post-hoc comparisons. The Friedman test reveals a significant difference across conditions ($\chi^2(6) = 162.36$, $p < .001$, $W = 0.576$). Post-hoc comparisons show that AURORA achieves significantly higher presence than the baseline ($Z = 5.97$, $p < .001$, $r = 0.87$) and all four static techniques (all $p < .001$, $r = 0.81$ – 0.86), and AURORA-HIL further improves presence over AURORA ($Z = 2.42$, $p = .016$, $r = 0.35$). These findings strongly support **H2**, indicating that the proposed framework provides a higher sense of presence and a better overall UX than conventional methods.

Post-task Interview Phase: We also report additional user study results evaluating participants' perceptions of the proposed AURORA framework compared with conventional methods. During the study, participants rated their experiences on a 5-point Likert scale [56], comparing AURORA with conventional methods without RL-based adaptation, where the applied visual techniques remained static, following prior work [9], [24]. Specifically, participants were asked to indicate their agreement with the following statements: **(S1)** *AURORA better balanced my discomfort, cognitive demand, and working memory than the conventional methods*, **(S2)** *The visual technique selection made by AURORA felt reliable and appropriate based on my VR experience*, **(S3)** *The automatic switching*

TABLE VI

PAIRWISE WILCOXON SIGNED-RANK COMPARISONS WITH HOLM-BONFERRONI CORRECTION, WHERE AR = AURORA, W = TEST STATISTIC, Z = STANDARDIZED WILCOXON STATISTIC, p = RAW p -VALUE, p (ADJ.) = HOLM-BONFERRONI ADJUSTED p -VALUE, AND r IS THE STANDARDIZED WILCOXON EFFECT SIZE. THE SIGN OF Z INDICATES THE DIRECTION OF THE EFFECT.

Condition Pair	FMS					CPL					CML					WM				
	W	Z	p	p (adj.)	r	W	Z	p	p (adj.)	r	W	Z	p	p (adj.)	r	W	Z	p	p (adj.)	r
Baseline vs. DFOV	34	5.603	<.001	<.001	0.817	237	-3.310	.001	.009	0.488	106	4.742	<.001	<.001	0.699	317	-2.614	.009	.088	0.381
Baseline vs. DGB	48	5.461	<.001	<.001	0.796	13	5.825	<.001	<.001	0.850	375	-2.000	.037	.188	0.292	496	-0.714	.179	.537	0.104
Baseline vs. DOF	22	5.730	<.001	<.001	0.836	278	3.021	.002	.017	0.441	368	-2.074	.037	.188	0.303	399	-1.541	.122	.488	0.227
Baseline vs. RF	168	4.191	<.001	<.001	0.611	387	-1.873	.060	.181	0.273	526	0.397	.691	1.000	0.058	375	1.808	.070	.349	0.267
Baseline vs. CR	12	5.836	<.001	<.001	0.851	315	2.635	.008	.050	0.384	152	4.355	<.001	<.001	0.635	251	-3.163	.002	.020	0.466
Baseline vs. AR-HIL	5	5.910	<.001	<.001	0.862	105	4.852	<.001	<.001	0.708	118	4.720	<.001	<.001	0.688	226	-3.572	<.001	.006	0.521
DFOV vs. DGB	331	-2.460	.013	.027	0.359	0	5.963	<.001	<.001	0.870	58	-3.355	<.001	<.001	0.781	319	2.587	.010	.088	0.377
DFOV vs. DOF	278	3.021	.003	.007	0.441	101	4.894	<.001	<.001	0.714	70	-5.223	<.001	<.001	0.762	437	1.339	.181	.537	0.195
DFOV vs. RF	104	-4.863	<.001	<.001	0.709	356	2.196	.028	.111	0.320	199	-3.857	<.001	.001	0.563	138	4.503	<.001	<.001	0.657
DFOV vs. CR	29	5.583	<.001	<.001	0.823	212	3.725	<.001	.002	0.543	341	2.360	.018	.108	0.344	380	-1.942	.051	.309	0.283
DFOV vs. AR-HIL	10	5.857	<.001	<.001	0.854	48	5.460	<.001	<.001	0.796	259	3.222	.001	.011	0.470	316	-2.453	.014	.112	0.362
DGB vs. DOF	148	4.402	<.001	<.001	0.642	80	-5.117	<.001	<.001	0.746	560	0.042	.966	1.000	0.006	451	-1.196	.232	.537	0.174
DGB vs. RF	228	-3.556	<.001	.001	0.519	11	-5.847	<.001	<.001	0.853	323	2.545	.011	.086	0.371	327	2.333	.019	.136	0.344
DGB vs. CR	32	5.624	<.001	<.001	0.820	333	-2.439	.015	.073	0.356	102	4.785	<.001	<.001	0.706	284	-2.958	.003	.037	0.431
DGB vs. AR-HIL	9	5.868	<.001	<.001	0.856	486	0.820	.412	.515	0.120	71	5.212	<.001	<.001	0.760	234	-3.487	<.001	.007	0.509
DOF vs. RF	37	-5.572	<.001	<.001	0.813	210	-3.746	<.001	.002	0.546	328	2.492	.012	.088	0.364	194	3.910	<.001	.002	0.570
DOF vs. CR	45	5.487	<.001	<.001	0.800	457	1.127	.260	.515	0.164	106	4.841	<.001	<.001	0.706	303	-2.762	.006	.062	0.403
DOF vs. AR-HIL	15	5.804	<.001	<.001	0.847	191	3.942	<.001	.001	0.575	60	5.328	<.001	<.001	0.777	247	-3.355	<.001	.011	0.489
RF vs. CR	13	5.825	<.001	<.001	0.850	249	3.185	.001	.011	0.470	201	3.836	<.001	<.001	0.560	180	-4.058	<.001	<.001	0.592
RF vs. AR-HIL	5	5.910	<.001	<.001	0.862	64	5.286	<.001	<.001	0.771	121	4.683	<.001	<.001	0.683	150	-4.376	<.001	<.001	0.638
CR vs. AR-HIL	375	1.995	.045	.045	0.291	252	3.302	<.001	.009	0.482	450	0.983	.323	.968	0.145	47	-5.467	<.001	<.001	0.797

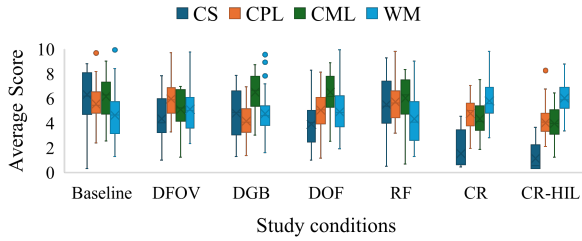


Fig. 7. Average scores (CS,CPL,CML,WM) across study conditions.

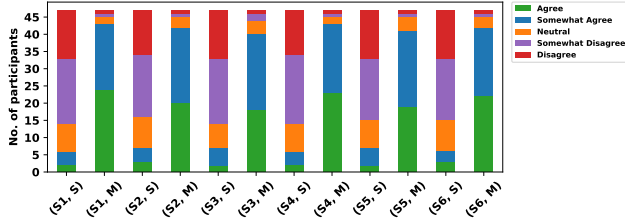


Fig. 8. User feedback in Likert scale between the AURORA framework denoted as (M), and without the AURORA framework denoted as the conventional (S) technique, in which the visual technique is static.

between different visual techniques in AURORA felt smooth and non-disruptive, (S4) The personalized version of AURORA felt more tailored to my needs than the non-personalized version, (S5) Using AURORA improved my overall VR experience compared to sessions with conventional methods, and (S6) I would recommend AURORA to others because of its adaptive approach compared to conventional methods. We select these six statements to evaluate the proposed AURORA framework because they collectively capture the key aspects of user experience and system effectiveness needed to assess a new framework, following prior work [9], [24]. To evaluate the rated users' experiences, we compute the mean and SD of the Likert score for each statement across all participants. The user feedback results are shown in Figure 8. We observe that approximately 89% of participants agreed or strongly agreed with all six statements, confirming that AURORA was perceived as more effective at reducing cybersickness, balancing cognitive load, improving working memory, and providing a more immersive experience than conventional methods. At the same time, most participants reported that

the conventional methods introduced unnecessary cognitive load and disrupted working memory during immersion. These findings further support H1, H2, and H3, indicating that AURORA is perceived as more effective across the targeted user cognitive states, more immersive, and more beneficial when personalized than conventional methods.

SSQ Results for All Study Conditions: In examining the SSQ scores, we compare the results from the pre-study and the post-study conditions for all participants and find a significant difference. For this, we first perform a Shapiro-Wilk test on these data, indicating that we cannot assume normality. Therefore, we conduct the Wilcoxon signed-rank test to quantify the difference between the pre and post-session SSQ subscores and the total scores. All subscores except Disorientation showed a significant post-session increase: Nausea ($Z = -4.12, W = 142, p = 0.003$), Oculomotor ($Z = -3.68, W = 118, p = 0.002$), Disorientation ($Z = -1.92, W = 185, p = 0.076$), and Total Score ($Z = -3.85, W = 105, p = 0.001$).

VII. LIMITATIONS AND FUTURE WORK

The AURORA framework has a few limitations. First, our user study was conducted with a specific participant group in a VR maze navigation task, and we did not explicitly examine how factors such as age, cultural background, or task diversity may influence cybersickness, cognitive load, and working memory [11]. Future work should therefore evaluate AURORA with more diverse participants, broader VR tasks, and longer exposure durations. Second, the MTL-enabled DL model used to predict user states for RL reward shaping is primarily trained on maze-walking data, and the RL policy is also developed in a VR maze simulation. Although domain randomization is applied across varied procedural settings during RL training, the framework has not yet been evaluated beyond this environment. Future work should extend training to multiple environments, evaluate performance on unseen scenarios, and assess AURORA in more diverse VR contexts. Finally, we plan to expand the visual technique library to include additional techniques (e.g., snap turning,

color manipulation, etc.) and evaluate the framework across diverse VR environments.

VIII. CONCLUSION

This work proposed AURORA, a multi-objective RL framework that jointly optimized users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory capabilities, thereby enhancing UX and comfort during VR immersion. Specifically, the framework incorporates an MTL-enabled DL model that predicts user cognitive states in real time from streaming eye- and head-tracking data, which are then used to construct a multi-objective reward function for RL-based policy learning. Based on this reward, the RL agent dynamically selects visual techniques from our developed visual technique library (DFOV, DGB, DOF, RF) and adaptively adjusts their intensity, enabling context-aware policies that optimize overall UX rather than focusing solely on cybersickness reduction. Furthermore, a HIL fine-tuning method further personalized the learned policy using in-session comfort ratings and post-session feedback, enabling adaptation to individual tolerance profiles. To validate the framework, we implemented AURORA on a consumer-grade VR headset (HTC Vive Pro Eye), released it as open source, and evaluated it through a user study with 47 participants. Results show that the proposed AURORA significantly reduces cybersickness while balancing cognitive load and improving working memory compared to static techniques, with approximately 89% of participants reporting it as more effective, less demanding, and more immersive. These findings demonstrate that multi-objective RL is a promising approach for jointly optimizing users' cognitive states in consumer VR headsets.

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